Dylan C. Meeks

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Research Interests

Computer Architecture; Heterogeneous Systems; Memory & Storage Systems; Processing-in-Memory; Operating Systems; Programming Models

Education

University of Illinois Urbana-Champaign

Ph.D. Computer Science — *August 2026 — Present*

- I work with Prof. Saugata Ghose at the University of Illinois focusing on Computer Architecture for emerging domains such as Processing-Using-Memory

University of Texas at Austin

B.S. Electrical and Computer Engineering, Minor: Chinese Language — *August 2022 — May 2026*

- Cumulative GPA: 3.7/4.0
- Relevant Coursework: Computer Architecture, Compilers, Operating Systems, Hardware Design Languages, Digital Logic Design, Programming Paradigms, Embedded Systems Laboratory, Algorithms, Probability and Random Processes

Work Experience

Teaching Assistant, CS 233 — August 2026 — December 2026

University of Illinois, Urbana-Champaign

- Led development of SPIMbot, the course's semester-long programming competition, directing a team of 4 undergraduate TAs through game design, implementation, testing, and release to 250+ students
- Enforced academic integrity on labs and assignments, reviewing submissions, and reporting cases to the instructor
- Supported accessibility efforts by tracking course materials that needed improvement and helping make content accessible to all students
- Provided weekly student support through 4 hours of office hours and the course Discord, helping students debug code and understand key topics, e.g., MIPS assembly, digital logic, verilog, etc.

Research Intern, Arcana Research Lab — May 2025 — August 2025

University of Illinois, Urbana-Champaign

- Built 3D transient heat models in **C++** for resistive memory devices; solved the heat equation to characterize thermal coupling and endurance limits
- Implemented Processing-Using-Memory (PUM) kernels to evaluate energy/thermal profile in a state-of-the-art simulator
- Designed and ran simulation test benches for massively parallel workloads, profiling power, latency, and scalability for general-purpose compute use cases
- Presented poster and PowerPoint presentation with results of research to peers and industry sponsors

Research Intern, CORSA Research Lab — May 2024 — August 2024

University of California, Irvine

- Investigated Dynamic Partial Reconfiguration (DPR) for FPGA-based machine learning accelerators to reduce idle power and adapt to workload phases
- Prototyped a DPR-enabled accelerator pipeline (compute/reduction stages) and achieved up to ~50% power reduction under representative driving traces

Projects

Pintos Operating System — January 2025 — Present

- Worked in group of two to implement user process virtualisation, virtual memory, and file system
- Implemented a full virtual-memory subsystem with on-demand paging, LRU-approximate eviction, and stack-growth heuristics
- Added full system-call interface: argument passing, process exec/wait, file I/O, and robust user-memory access
- Re-engineered Pintos file system to support indexed/extensible files, hierarchical directories, and fine-grained synchronization

FPGA Projects Various January 2025 — Present

- Designed and implemented digital systems in **Verilog** on Xilinx FPGAs, including combinational/sequential circuits, FSMs, and memory-mapped I/O
- Built a Fitbit-style activity tracker and parking meter with real-time 7-segment display multiplexing, debouncing, and clock division
- Developed a stack calculator using Block RAM and tri-state bus interfaces; designed a single-cycle MIPS processor with full datapath and control unit

Roofline Analysis Tool for LLVM MLIR in Collaboration with Ericsson October 2025 — Present

- Lead 5 person team as to design a roofline analysis tool for MLIR code for performance analysis
- Implemented roofline analysis of input code using LLVM MLIR python bindings
- Collaborated in designing intuitive user interface inspired by CompilerExplorer to present results of the user

COOL Compiler August 2025 — December 2025

- Implemented recursive-descent parser with LL(1) lookahead and operator precedence handling; constructed AST using Cool support libraries with proper error reporting
- Designed and implemented code generation to LLVM IR for full COOL semantics: classes, inheritance, dynamic dispatch, vtables, SELF_TYPE, and runtime library integration with boxing/unboxing optimizations for primitive types
- Developed LLVM optimization passes: loop analysis (natural loop identification), loop-invariant code motion with alias analysis, and sparse conditional constant propagation (SCCP)
- Implemented a greedy local register allocator as an LLVM MachineFunctionPass, handling physical/virtual register mapping, spilling, reloading, subregisters, and function call clobbering across x86 and AArch64 backends

Cypher to SQL Database-as-a-Service August 2025 — December 2025

- Built a multi-paradigm database-as-a-service system with a Smalltalk client, **Go** REST server, and **Rust** query transformer, translating Cypher graph queries to **SQL** for relational backends
- Implemented a custom Cypher-to-SQL parser by extending Apache DataFusion's sqlparser-rs crate, adding grammar support and desugaring logic for graph pattern matching operations
- Designed modular architecture enabling pluggable database backends (SQLite/MySQL) and integrated Neo4j for query validation, with comprehensive testing and CI/CD pipeline

Wireless Electronic Battleship August 2025 — December 2025

- Designed and built multiple custom PCBs implementing a DAC and amplifier circuit for audio, a keyboard matrix for input, and using communication protocols such as UART, I2C, and SPI
- Created custom 3D printed case for the PCBs designing space for multiple switches and button
- Developed custom firmware to support multiplayer gameplay across multiple copies of the game

Performance Simulator for LC3b ISA August 2024 — December 2024

- Implemented two-pass assembler with symbol table resolution, parsing LC-3b assembly into machine code with support for labels, pseudo-ops, and all instruction variants
- Developed cycle-accurate simulator of LC-3b microarchitecture: datapath, microsequencer, memory subsystem, and control store
- Extended simulator with interrupt/exception handling (timer, protection, unaligned access, unknown opcode), nested interrupts, privilege mode switching, and RTI instruction
- Implemented virtual memory with one-level page table translation, page faults, reference/modified bits, and protection checks; designed pipelined 5-stage processor with hazard detection and stall logic

Skills

- **Programming**: C/C++, Rust, Go, Verilog, Python, SQL, ARM Assembly, Git, Matlab
- **Modeling and Design**: Vivado, Vivado HLS, Vitis, KiCAD, SolidWorks, FreeCAD, LTSpice
- **Hardware**: AMD Kria KV260 Vision AI Starter Kit FPGA, Diligent Basys 3 FPGA, Ti TM4C, Raspberry Pi